

Carl A. Schmidt

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Appointments

Research Assistant Professor, Boston University	2021 - Present
Research Scientist, Boston University	2017 - 2021
Research Associate, CNRS/LATMOS, Sorbonne Univ. (F. Leblanc Supervisor)	2015 - 2017
Research Associate, Univ. of Virginia (R. E. Johnson Supervisor)	2013 - 2015
Graduate Research Assistant, Boston Univ. (M. Mendillo Supervisor)	2006 - 2013
Undergraduate Research Assistant, Univ. Colorado (F. Hearty Supervisor)	2002 - 2005

Research Areas: Planetary aeronomy, plasma interactions with surfaces and atmospheres, Monte-Carlo modelling, telescope-based observation and instrumentation

Education

Ph.D., Astronomy, Boston University	2013
Thesis: <i>Mercury's Sodium Exosphere</i> (M. Mendillo Advisor)	
M.A., Astronomy, Boston University	2008
B.A., Physics, University of Colorado	2005

Teaching Experience

Instructor, Boston Univ. 2022 (Fall), 2023 (Fall & Spring), 2024 (Fall), 2025 (Spring)

- *AS865 & AS866 Graduate Research Seminar I & II* Weekly seminar offering astronomy graduate students the skills and practice needed for oral presentations on current research topics and to receive peer and expert feedback, 2 credits

Guest Lecturer, Boston Univ. 2022 (Fall term)

- *CC111 Core Natural Science I: Origins- of the Big Bang, Earth, Life and Humanity* 5 lectures in a team-taught course in Boston University's Core Curriculum program. Targeted at incoming students fulfilling scientific inquiry, quantitative reasoning and teamwork/collaboration requirements, 4 credits, expected enrollement is roughly 100 students.

Instructor, Boston Univ. 2018 - 2021 (Summer term, annually)

- *AS101 The Solar System* course in Boston University's Astronomy Dept targeted at undergraduate non-majors fulfilling a laboratory science requirement, 4 credits, typically 16 students enrolled with 1 teaching assistant.

Teaching Assistant, Boston Univ.

2007 (Spring term)

- Lab instructor for *AS101 The Solar System* undergraduate course

Undergraduate Research Advisor:

- Chase Young (Spring - Summer 2018)
- Mikhail Sharov (Fall 2018 - Spring 2021)
- Cameron Moye (Univ. Maryland, NASA SUPPR intern, Summer 2020)
- Aishwarya Ganesh (Univ. Texas, NASA SUPPR intern, Summer 2021)
- Patrick Lierle (Summer 2019 - Summer 2022)
- Madison Jordan (North Carolina A&T, NSF PAARE intern, Summer 2024)

Graduate Research Advisor:

- Emma Lovett (Fall 2021 -), NASA FINESST Graduate Fellowship
- Patrick Lierle (Fall 2022 -)

Academic Service

PhD Dissertation Committees:

- Rozenn Robidel (PhD 2023, Université Paris-Saclay) *Study of the exosphere of Mercury with the PHEBUS spectrograph on the BepiColombo mission*
- Alec Daly (PhD candidate, Boston University) *TBD*
- Lou Baya Ould Rouis (PhD candidate, Boston University) *TBD*

Journal Articles Under Review (students underlined)

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- E. Lovett, **C. Schmidt**, P. R. Lierle (2025) *Europa's Sodium and Potassium Exosphere during Juno's Flyby*, Planetary Science Journal, *under review*
 - A. V. Oza, R. E. Johnson, **C. A. Schmidt**, W. Calvin (2025) *Plasma and Thermal Processing Leading to Spatial and Temporal Variability of the Trapped O₂ at Europa and Ganymede*, Astrobiology, *under review*
 - P. R. Lierle, E. Lovett, **C. Schmidt**, A. Merkel (2025) *Characterizing Dynamical Processes in Surface-Bound Exospheres Via Resolved Sodium D Emissions*, Frontiers in Astronomy and Space Sciences, *under review*
 - P. R. Lierle & **C. Schmidt** (2025) *Sodium and Potassium Linewidths as an Atmospheric Escape Diagnostic at Mercury*, Journal of Geophysical Research: Planets, *under review*

- Z. Milby, K. de Kleer, **C. Schmidt** (2025) *Electron Properties in Io's Ionosphere and Exosphere Revealed through Optical Auroral Emissions*, Planetary Science Journal, *under review*
- A. V. Oza, A. Gebek, M. Meyer zu Westram, A. Tokadjian, A. Piro, R. Hu, A. Unni, R. Chari, Aaron Bello-Arufe, **C. Schmidt**, A. Louca, Y. Miguel, R. Estrela, J. Yang, M. Damiano, Y. Hasegawa, L. Welbanks, D. Powell, Y. K. Yung (2025) *Volcanic Satellites Tidally Venting Na, K, SO₂ in Optical & Infrared Light*, Monthly Notices of the Royal Astronomical Society, *under review*
- I. de Pater, D. F. Strobel, A. G. Davies, J. Saur, L. Roth, K. de Kleer, E. Lellouch, Z. Milby, **C. Schmidt**, T. Fouchet, M. H. Wong, L. N. Fletcher, J. Harkett, M. Roman, R. J. Cartwright, J. E. Perry, J. Renaud-Kim, C. Jordan, S. Schlegel, D. A. Williams, D. M. Nelson (2024) *First Detection of [S I] in Near-IR JWST Observations of Io in Eclipse, and Comparison with SO Emissions, Evolving Volcanic Eruptions, and prior UV HST-STIS [S I] emissions*. Journal of Geophysical Research: Planets, *under review*

Peer-Reviewed Book Chapters (students underlined)

- **C. Schmidt** and J. Baumgardner (2022) *Lunar Atmosphere, Alkali Lunar Exosphere in Encyclopedia of Lunar Science*, Editors B. Cudnik & C. Ahrens, Springer, [DOI](#).
- F. Leblanc, **C. Schmidt**, V. Mangano, A. Mura, G. Cremonese, J. M. Raines, J.M. Jasinski, M. Sarantos, A. Milillo, R.M. Killen, S. Massetti, T. Cassidy, R.J. Vervack Jr., S. Kameda, M.T. Capria, M. Horanyi, D. Janches, A. Berezhnoy, A. Christou, T. Hirai, P. Lierle and J. Morgenthaler (2022) *Comparative Na and K Mercury and Moon exospheres in Surface Bounded Exospheres and Interactions in the Solar System*, Space Sciences Series of ISSI, Springer. Jointly published in Space Science Reviews, Vol 218, 2, [DOI](#).

Peer-Reviewed Journal Articles (students underlined)

- C. Nixon and 38 co-authors, including **C. Schmidt** (2025) *Titan's Atmosphere in Late Northern Summer from JWST and Keck Observations*. Nature Astronomy, *in press*
- A. Unni, A. V. Oza, S. Thirupathi, H. J. Hoeijmakers, J. V. Seidel, **C. A. Schmidt**, K. de Kleer, A. D. Baker, B. Manjunath, A. Gebek, M. Meyer zu Westram (2024) *Doppler Shifted Transient Sodium Detection by KECK/HIRES*, Monthly Notices of the Royal Astronomical Society: Letters, slaf031, [DOI](#).
- K. Roberts, L. Moore, J. O'Donoghue, H. Melin, T. Stallard, K. Knowles, **C. Schmidt**, P. Tiranti (2025) *Spatiotemporal Variations of Temperature in Jupiter's Upper Atmosphere*. Planetary Science Journal, Vol. 6, 92, [DOI](#).
- L. Roth, A. Blöcker, K. de Kleer, D. Goldstein, E. Lellouch, J. Saur, **C. Schmidt**, D.F. Strobel, C. Tao, F. Tsuchiya, V. Dols, H. Huybrighs, A. Mura, J. R. Szalay, S. V. Badman, I. de Pater, A.-C. Dott, M. Kagitani, L. Klaiber, R. Koga, A. McEwen,

- Z. Milby, K.D. Retherford, S. Schlegel, N. Thomas, W.L. Tseng, A. Vorburger (2025) *Mass Supply from Io to Jupiter's magnetosphere*. Space Science Reviews, 221, 13, [DOI](#), [arXiv](#).
- C. Gray, K. Peter, M. Pätzold, S. Tellmann, T. Nordheim, **C. Schmidt**, N Chanover and P. Withers (2025) *Venus' O 5577Å Oxygen Green Line: A Global Diffuse Proton-induced Aurora*. Journal of Geophysical Research: Space Physics, V130 (2), e2024JA032851, [DOI](#).
 - E. G. Nerney, F. Bagenal, **C. Schmidt** (2024) *Simulations of Optical Emissions in Io's Plasma Torus*. Journal of Geophysical Research: Space Physics, V130 (1), e2024JA033232, [DOI](#).
 - **C. Schmidt** (2024) *On the Implications of Ground-Based High-Definition Imaging of Io's Surface*. Geophysical Research Letters, V51, 21, e2024GL112169, [DOI](#).
 - A. V. Oza, J. V. Seidel, H. J. Hoeijmakers, A. Unni, A. Y. Kesseli, **C. Schmidt**, S. Thirupathi, A. Gebek, M. Meyer zu Westram, S. Sousa, A. Bello-Arufe, R. Lopes, R. Hu, C. Fisher, S. Charnoz, A. D. Baker, S. P. Halverson, K. de Kleer, N. Schneider, A. Psaridi, M. Lendl, A. Wyttenbach, I. Bhatnagar and R. E. Johnson (2024) *Redshifted Sodium Transient Near Exoplanet Transit*, Astrophysical Journal Letters, 973, L53, [DOI](#).
 - M. Yoneda, F. Tsuchiya, **C. Schmidt**, M. Kagitani and T. Sakanoi (2024) *Major Brightening Events in Jupiter's Sodium Nebula during Juno era*. Icarus, V425, [DOI](#).
 - Z. Milby, K. de Kleer, **C. Schmidt**, and F. Leblanc (2024) *Short-Timescale Spatial Variability of Ganymede's Optical Aurora*, Planetary Science Journal, 5, 153, [DOI](#).
 - J. Morgenthaler, J. Rathbun, **C. Schmidt**, J. Baumgardner and N. Schneider (2024) *Erratum: 'Large Volcanic Event on Io Inferred from Jovian Sodium Nebula Brightening' (ApJL 871:L23)*, Astrophysical Journal Letters, 966 (2), L40, [DOI](#).
 - J. Morgenthaler, **C. Schmidt**, M. F. Vogt, N. M. Schneider and M. Marconi (2024) *Jovian Sodium Nebula and Io plasma torus S⁺ and Brightnesses 2017 – 2023: Insights into Volcanic vs. Sublimation Supply*. Journal of Geophysical Research: Space Physics, 129, 3, [DOI](#).
 - D. Bhattacharyya, J. T. Clarke, M. Mayyasi, V. Shematovich, D. Bisikalo, J. Y. Chaufray, E. Thiemann, J. Halekas, **C. Schmidt**, J. L. Bertaux, M. S. Chaffin, and N. M. Schneider (2023) *Evidence of Hot Hydrogen in the Exosphere of Mars Resulting in Enhanced Water Loss*. Journal of Geophysical Research: Planets, 128, 8, [DOI](#).
 - P. Lierle, **C. Schmidt**, J. Baumgardner, L. Moore and E. Lovett (2023) *The Rapid Imaging Planetary Spectrograph*. Publications of the Astronomical Society of the Pacific, 135, 095002, [arXiv](#), [DOI](#)
 - Q. Zhang, K. Battams, Q. Ye, M. Knight and **C. Schmidt** (2023) *Sodium Brightening of (3200) Phaethon Near Perihelion*. The Planetary Science Journal, Vol. 4, 70, [DOI](#).
 - K. de Kleer, Z. Milby, **C. Schmidt**, M. Camarca and M. Brown (2023) *The Optical and Near-Infrared Aurorae of Europa, Ganymede and Callisto*. The Planetary Science Journal, Vol. 4, 37, [DOI](#).

- **C. Schmidt**, M. Sharov, K. de Kleer, N. Schneider, I. de Pater, P.H. Phipps, A. Conrad, L. Moore, P. Withers, J. Spencer, J. Morgenthaler, I. Ilyin, K. Strassmeier, C. Veillet, J. Hill, and M. Brown (2023) *Io's Optical Aurorae in Jupiter's Shadow*. The Planetary Science Journal, Vol. 4, 36, [DOI](#).
- P. R. Lierle, **C. Schmidt**, J. Baumgardner, L. Moore, T. Bida and R. Swindle (2022) *The Spatial Distribution and Temperature of Mercury's Potassium Exosphere*. The Planetary Science Journal, Vol. 3, 87, [DOI](#).
- **C. Schmidt** (2022) *Doppler-Shifted Alkali D Absorption as Indirect Evidence for Exomoons*. Frontiers in Astronomy and Space Sciences, Vol. 9, 801873, [DOI](#).
- A. L. E. Werner, S. Aizawa, F. Leblanc, J. Y. Chaufray, R. Modolo, J. M. Raines, W. Exner, U. Motschmann and **C. Schmidt** (2022) *Ion density and phase space density distribution of planetary ions Na^+ , O^+ and He^+ in Mercury's magnetosphere*. Icarus, Vol. 372, 114734, [DOI](#).
- T. Cassidy, **C. Schmidt**, A. Merkel, J. Jasinski and M. Burger (2021) *Detection of Large Exospheric Enhancements at Mercury due to Meteoroid Impacts*, The Planetary Science Journal, Vol. 2, 175, [DOI](#).
- J. Baumgardner, S. Luettggen, **C. Schmidt**, M. Mayyasi, S. Smith, C. Martinis, J. Wroten, L. Moore and M. Mendillo (2021) *Long-Term Observations and Physical Processes in the Moon's Extended Sodium Tail*, Journal of Geophysical Research: Planets, Vol. 126, 3, [DOI](#).
- V. Mangano and 61 co-authors including C. Schmidt (2021) *BepiColombo science investigations during cruise and flybys at the Earth, Venus and Mercury*, Space Science Reviews, Vol. 217, 23, [DOI](#).
- **C. Schmidt**, J. Baumgardner, L. Moore, T. A. Bida, R. Swindle and P. Lierle (2020) *The Rapid Imaging Planetary Spectrograph: Observations of Mercury's Sodium Exosphere in Twilight*. The Planetary Science Journal, Vol. 1, 4, [DOI](#).
- A. Oza, R.E. Johnson, E. Lellouch, **C. Schmidt**, N. Schneider, C. Huang, D. Gamborino, A. Gebek, A. Wyttenbach and B-O Demory. (2019) *Sodium and Potassium as Remnants of Volcanic Satellites Orbiting Close-in Gas Giant Exoplanets*, Astrophysical Journal, V885, 2, [DOI](#).
- L. Moore, H. Melin, T. Stallard, J. O'Donoghue, J. Moses, S. Miller and **C. Schmidt** (2019) *Modelling H_3^+ in Planetary Atmospheres: Effects of Vertical Gradients on Observed Quantities*, Philosophical Transactions of the Royal Society A, 377, [DOI](#).
- R.E. Johnson, A. Oza, F. Leblanc, **C. Schmidt** and T.A. Nordheim (2019) *The Origin and Fate of O_2 in Europa's Ice: An Atmospheric Perspective*. Space Science Reviews, 215 (1), 20, [DOI](#).
- J. Morgenthaler, J. Rathbun, **C. Schmidt**, J. Baumgardner and N. Schneider (2019) *Large Volcanic Event on Io Inferred from Jovian Sodium Nebula Brightening*, Astrophysical Journal Letters, 871 (2), L23, [DOI](#).
- A. Oza, F. Leblanc, R. E. Johnson, **C. Schmidt**, L. Leclercq, T. Cassidy and J.-Y. Chaufray (2019) *Dusk Over Dawn O_2 Asymmetry in Europa's Near-Surface Atmosphere*. Planetary & Space Science, 167, 23-32, [DOI](#).

- **C. Schmidt**, N. Schneider, F. Leblanc, C. Gray, J. Morgenthaler, J. Turner and C. Grava (2018) *A Survey of Visible S⁺ Emission in Io's Plasma Torus During the Hisaki Epoch*. Journal of Geophysical Research: Space Physics, 123, 7, 5610-5624, [DOI](#).
- F. Leblanc, A. Oza, L. Leclercq, **C. Schmidt**, T. Cassidy, R. Modolo, J.Y. Chaufray and R. E. Johnson (2017) *On the Orbital Variability of Ganymede's Atmosphere*. Icarus, Vol. 293, p. 185-198, [DOI](#).
- J. D. Turner, D. Christie, P. Arras, R. E. Johnson and **C. Schmidt** (2016) *Investigation of the environment around close-in transiting exoplanets using CLOUDY*. Monthly Notices of the Royal Astronomical Society, Vol 458 (4), p.3880-3891, [DOI](#).
- **C. Schmidt** (2016) *High Resolution Integral-Field Spectroscopy of Gas and Ion Distributions in the Coma of Comet C/2012 S1 ISON*. Icarus, Vol 265, p. 35-41, [DOI](#).
- R.E. Johnson, A. Oza, L.A. Young, A.N. Volkov and **C. Schmidt** (2015) *Volatile Loss and Classification of Kuiper Belt Objects*. Astrophysical Journal, Vol 809 (1), 43, [DOI](#).
- N.-E. Raouafi, C. M. Lisse, G. Stenborg, G. H. Jones and **C. Schmidt** (2015) *Dynamics of HVEC's emitted from comet C/2011 L4 as observed by STEREO*. Journal of Geophysical Research, Vol 120 (7), pp. 5329-5340, [DOI](#).
- **C. Schmidt**, R.E. Johnson, J. Baumgardner and M. Mendillo (2015) *Observations of Sodium in the Coma of Comet C/2012 S1 (ISON) During Outburst*. Icarus, Vol 247, p. 313-318, [DOI](#).
- **C. Schmidt** (2013) *Monte-Carlo Modeling of North-South Asymmetries in Mercury's Sodium Exosphere*, Journal of Geophysical Research, Vol 118, A50396, [DOI](#).
- **C. Schmidt**, J. Baumgardner, M. Mendillo and J. Wilson (2012) *Escape rates and variability constraints for high-energy sodium sources at Mercury*, Journal of Geophysical Research, Vol 117, A03301, [DOI](#).
- **C. Schmidt**, J. Wilson, J. Baumgardner and M. Mendillo (2010) *Orbital Effects on Mercury's Escaping Sodium Exosphere*, Icarus, Vol 207 (1), p. 9-16, [DOI](#).

Conference Proceedings and Abstracts (students underlined)

- M. Yoneda, F. Tsuchiya, C. Schmidt, J. Morgenthaler, M. Kagitani, T. Sakanao (2025) *Variations in Jupiter's extended sodium nebula over two decades*, European Geophysical Union Meeting, held 27 April - 2 May in Vienna, Austria.
- Lierle, P. & Schmidt, C (2025) *Na & K Linewidths as a Proxy for Exospheric Escape at Mercury*, Mercury Exploration Assessment Group, held Feb 4-6 in Laurel, MD.
- Vervack, R. J., Killen, R. M., Schmidt, C. A., Merkel, A. W., Burger, M. H. (2024) *Mercury's Exosphere: A Spatial and Temporal Investigation of Less Abundant and Weakly Emitting Species*, American Geophysical Union, Fall Meeting.
- Lierle, P. & Schmidt, C., Gray, C., Withers, P. (2024) *Red and Green Line Oxygen Emission in the Venusian Atmosphere*, American Geophysical Union, Fall Meeting.
- Schmidt, C. (2024) *Chlorine in the Io plasma torus*, American Geophysical Union, Fall Meeting 2024.

- Roberts, K. et al. (2024) *High-Resolution Maps of Magnetically-Organized Temperature and Density Structures in Jupiter's Upper Atmosphere*, American Geophysical Union, Fall Meeting.
- Morgenthaler J., Schmidt, C., Vogt, M., Schneider, N., Rutala (2024) *Long-term modulation in the position of the Io plasma torus: challenging the standard convection electric field picture*, American Geophysical Union, Fall Meeting.
- Schmidt, C., Merkel, A., Vervack, R., Lierle, P. (2024) *Understanding Seasonal Modulations in MESSENGER's Observations of Mercury's Sodium Exosphere*, Fall Meeting.
- Gray, C., Schmidt, C., Lierle, P. (2024) *The Behavior of the O I 557.7 Oxygen Green Line and Discovery of CO B-A 483.5 nm Aurora on Venus*, Venus Exploration and Analysis Group (VEXAG) Meeting, held 17-19 November in Louisville, KY, 3066, id.8020
- Lovett, E., Schmidt, C., Lierle, P. (2024) *Mapping Europa's Alkali Exosphere During Juno's 2022 Flyby*, EPSC2024-658
- Lierle, P., Schmidt, C. (2024) *Linewidth Measurements of Mercury's Alkali Exosphere*, EPSC2024-757
- Roth, L., Schmidt, C., Saur, J., Molyneux, P., Retherford, K., Mykola, I. (2024) *Ganymede's oxygen and water atmosphere during eclipse passage*. Outer planet moon-magnetosphere interaction workshop held May 13-17 in Dublin, IRL.
- Nixon, C. et al. (2023) *Observations of Titan in 2022-2023 with JWST and Keck*, American Geophysical Union, Fall Meeting 2023, abstract #P32B-06.
- Lovett, E., Schmidt, C., Lierle, P. (2023) *Europa's Alkali Exosphere During the 2022 Juno Flyby*, American Geophysical Union, Fall Meeting, abstract #P42A-07.
- Schmidt, C., Spencer, J., Roth, L., Ganesh, A., Moye, C., Saur, J., Retherford, K., Grodent, D. (2023) *Io's Far Ultraviolet Emissions Surrounding its Passage through Jupiter's Shadow*, American Geophysical Union, Fall Meeting 2023.
- de Kleer, K., Milby, M., Schmidt, C., Camarca, M., Brown, M. (2023) *The Optical Aurorae of Europa, Ganymede, and Callisto*, American Geophysical Union, Fall Meeting 2023, abstract #P42A-05.
- Morgenthaler J., Schmidt, C., Vogt, M., Schneider, N. (2023) *Variation in the brightness of the Jovian sodium nebula and Io plasma torus since 2017: a tool for studying gas-producing geologic processes on Io*, American Geophysical Union, Fall Meeting 2023, abstract #P33C-3170
- Schmidt, C., Baumgardner, J., Luetngen, S., Mayyasi, M., Smith, S. M., Martinis, C. R., Wroten, J., Moore, L., Mendillo, M. (2023) *Characterizing the Lunar Sodium Exotail Using Earth as a Lens*, American Geophysical Union, Fall Meeting 2023, abstract #P21C-3032.
- Morgenthaler J., Schmidt, C., Vogt, M., Schneider, N. (2023) *Io plasma torus observations on 550 nights during the Juno era: A comprehensive record of the modulation in the flow of plasma into and out of Jupiter's magnetosphere*, American Geophysical Union, Fall Meeting 2023, abstract #SM21A-07.

- Nerney, E., Bagenal, F., Wilson, R. J., Schmidt, C. (2023) A 3D Model of the Io Plasma torus and Model Comparisons with Observations, American Geophysical Union, Fall Meeting 2023, abstract #P03-06A.
- Lovett, E., Schmidt, C. (2023) *Mapping Europa's Salty Atmosphere*, Workshop on the Origins and Habitability of the Galilean Moons, held 24-26 October in Marseille, France.
- Morgenthaler J., Schmidt, C., Vogt, M., Schneider, N. (2023) Io plasma torus S+ and Jovian sodium nebula brightnesses since 2017: context for Io volcanic, atmospheric, and Jovian magnetospheric studies, EPSC-DPS Joint Meeting, held 1-6 Oct in San Antonio, Texas.
- Nerney, E., Bagenal, F., Schmidt, C. (2023) *A 3D Model of the Io Plasma torus and Model Comparisons with Observations*, EPSC-DPS Joint Meeting, held 1-6 Oct in San Antonio, Texas.
- Milby, Z. de Kleer, K., Schmidt, C. (2023) *Short-Timescale Variability of Ganymede's Optical Aurora*, EPSC-DPS Joint Meeting, held 1-6 October in San Antonio, Texas.
- Roberts, K., Moore, L., Stallard, T., Chowdhury, N., Melin, H., O'Donoghue, J., Vogt, M., Mohamed, K., Agiwal, O. (2023) *Mapping Temporal and Spatial Temperature Variations in Jupiter's Upper Atmosphere*, EPSC-DPS Joint Meeting, held 1-6 Oct in San Antonio, Texas.
- Lovett, E., Schmidt, C., Lierle, P. (2023) *Europa's Alkali Exosphere During the 2022 Juno Flyby*, EPSC-DPS Joint Meeting, held 1-6 Oct in San Antonio, Texas.
- Lierle, P., Schmidt, C., Lovett, E. (2023) *Measurements of Heating in Mercury's Alkali Exotail*, EPSC-DPS Joint Meeting, held 1-6 Oct in San Antonio, Texas.
- Schmidt, C., Merkel, A., Vervack, R. (2023) *Exospheric Redistribution of Elements in Mercury's Soil*, EPSC-DPS Joint Meeting, held 1-6 Oct in San Antonio, Texas.
- Nixon C. A. and 36 co-authors including Schmidt, C. (2023) *Observations of Clouds on Titan with JWST/NIRCam and Keck/NIRC-2*, European Geophysical Union Meeting, held 23–28 April in Vienna, Austria.
- Clarke, J. T., Bhattacharyya, D., Mayyasi, M., Shematovich, V., Bisikalo, D., Chaufray, J. Y., Thiemann, E., Halekas, J., Schmidt, C., Bertaux, J. L., Chaffin, M. S. and Schneider, N. M. (2023) *Evidence of Hot Hydrogen in the Exosphere of Mars Resulting in Enhanced Water Loss*. European Geophysical Union Meeting, held 23–28 April in Vienna, Austria.
- Cartwright, R., de Kleer, K., Schmidt, C. A., Villanueva, G. L., Beddingfield, C. B., Nordheim, T. A., Hand, K. P., Glein, C. R., Emery J. P., Hanley, J. and Thieberger, C. (2023) *Investigating the Nature and Origin of Hydrated Salts on Europa*. 54th Lunar and Planetary Science Conference, held March 13–17 in The Woodlands, TX, [link](#).
- Nixon C. A. and 36 co-authors including Schmidt, C. (2022) *First Observations of Titan with the James Webb Space Telescope*. American Geophysical Union Fall Meeting held 12-16 Dec. in Chicago, IL, abstract #P52D-1572.
- Morgenthaler J., Schmidt, C., Marconi, M., Vogt, M., Schneider, N. (2022) Find Your Favorite Io Volcanic Enhancement! A Global View of the Jovian Magnetosphere During

the Juno Mission as Recorded by PSI’s Io Input/Output Observatory (IoIO). American Geophysical Union Fall Meeting held 12-16 Dec. in Chicago, IL, abstract #SM42F-2232.

- Solorio, W., Merkel, A. W., Brain, D. A., Schmidt, C. (2022) Understanding the Role of Mercury in Investigating the Connection Between Planetary Habitability, Atmospheres, and Magnetic Fields. American Geophysical Union Fall Meeting held 12-16 Dec. in Chicago, IL, abstract #P42E-2442.
- Schmidt, C. (2022) Na and K Absorption in Solar System Transit Spectroscopy, Exoplanets in Our Backyard 2.0, held 2-4 Nov in Albuquerque, NM, LPI 2687, 3033 [link](#).
- Milby, Z., de Kleer, K., Schmidt, C., Camarca, M., Brown, M. (2022) Discovery of New Oxygen Aurora on Europa, Ganymede and Callisto. American Astronomical Society, DPS meeting #54, held 2-7 October in London, Ontario, id.513.08
- de Kleer, K., Milby, M., Schmidt, C., Camarca, M., Brown, M. (2022) The Red/Green Oxygen Aurorae of Europa and Ganymede. American Astronomical Society, DPS meeting #54, held 2-7 October in London, Ontario, id.513.09
- Moore, L., Stallard, T., O’Donoghue, J., Melin, H., Chowdhury, N., Johnson, R., Vogt, M., Schmidt, C. and Orton, G. (2022) Ionospheric temperature variability above Jupiter’s Great Red Spot, EPSC2022-564. EPSC Meeting, held 18-23 Sept 2022 in Granada, Spain, [link](#).
- Morgenthaler J., Marconi, M., Schmidt, C., Vogt, M., Schneider, N. (2022) Using the Io plasma torus as a probe of Io’s exosphere, Magnetospheres of the Outer Planets, held July 11-15, 2022 in Liege, BE, [link](#).
- Morgenthaler J., Marconi, M., Schmidt, C., Vogt, M., Schneider, N. (2022) Using Io Input/Output observatory (IoIO) observations to determine if mass flow in Jupiter’s magnetosphere driven by internal or external processes, Magnetospheres of the Outer Planets, held July 11-15, 2022 in Liege, BE, [link](#).
- Schmidt, C., Cassidy, T., Merkel, A., Jasinski, J., Burger, M. (2022) Impact Events Observed by MESSENGER UVVS, Mercury 2022: Current and future science of the innermost planet, held 7-10 June 2022 in Orléans, France. [link](#).
- Mangano, V., Schmidt, C., Vervack, R., Morgenthaler, J., Leblanc, F., Lierle, P., Del Moro, D. (2022) Coordinated Campaign of Ground-Based Observations of Mercury’s Exosphere in 2021, Mercury 2022: Current and future science of the innermost planet, held 7-10 June 2022 in Orléans, France. [link](#).
- Oza, A., Gebek, A., Seidel, J., Hoeijmakers, J., Unni, A., Thirupathi, S., Schmidt, C., Baker, A., de Kleer, K., Lopes, R., Johnson, R. E. (2022) Transient Sodium and Potassium Clouds at Candidate Exomoon Systems, Exoplanets IV, held 1-6 May in Las Vegas, NV, Bulletin of the American Astronomical Society, Vol. 54 (5), [link](#).
- Schmidt, C., Lierle, P., Mangano, V., Leblanc, F., Morgenthaler, J., Vervack, R. (2022) Coordinated Ground-Based Measurements of Mercury’s Exosphere, Mercury Exploration Assessment Group, held Feb 1–3, 2022, Virtual, [link](#).

- Schmidt, C. (2022) UVVS Measurements of Impactor Plumes: New Insights and Open Questions, Mercury’s Surface Response to the Interplanetary Environment: Identifying Needed Studies in Laboratory Astrophysics, held Jan 24–27, 2022, Virtual, [link](#).
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- Schmidt C., Baumgardner, J., Mendillo, M., Wilson J. K. (2011), Escape rates and variability constraints for high-energy sodium sources at Mercury, EPSC-DPS Joint Meeting 2011, held 2-7 Oct 2011 in Nantes, France, Vol. 6, EPSC-DPS2011-100.
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- Schmidt, Wilson, Baumgardner, J., Mendillo (2009) Variability in Mercury’s Escaping Sodium Atmosphere, American Astronomical Society, DPS meeting #41, id.35.01
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- Schmidt, Wilson, Baumgardner, J., Mendillo (2008) Wide Field Observations of Mercury’s Extended Sodium Exosphere, 37th COSPAR Scientific Assembly, held 13-20 July 2008 in Montreal, Canada, p.2775, B07-0036-08
- Schmidt, Baumgardner (2007) Boston University Calibration Facility for Optical Aeronomy. CEDAR Meeting Abstract

Non-Peer-Reviewed Publications

- Chanover, N., Schmidt, C., & DeColibus, D. (2021) *The Continued Relevance of 4m Class Telescopes to Planetary Science in the 2020s* White paper #497 submitted to the Decadal Survey in Planetary Science and Astrobiology 2023-2032, Bulletin of the AAS, 53(4), [DOI](#).
- P. Prem, A. Kereszturi, A. Deutsch, C. Hibbitts, C. Schmidt and 36 co-authors (2021) *Lunar Volatiles and Solar System Science*, White paper #68 submitted to the Decadal Survey in Planetary Science and Astrobiology 2023-2032, Bulletin of the AAS, 53(4), [ArXiv](#), [DOI](#).
- A. Deutsch, N. Chabot, A. Maiti, A. Luspai-Kuti, A. Kereszturi, A. Lucchetti, A. Virkki, A. Colaprete, A. Vorburger, B. Byron, B. Jones, B. Anzures, B. Butler, C. Schmidt and 59 co-authors (2021) *Science Opportunities offered by Mercury’s Ice-Bearing Polar Deposits*. Whitepaper #69 submitted to the Planetary Science and Astrobiology Decadal Survey 2023-2032, Bulletin of the AAS, 53(4), [DOI](#).
- J. Clarke, C. Schmidt, J. Baumgardner, C. Carveth, M. Matta, M. Mendillo, L. Moore, and P. Withers (2013) White Paper on Comparative Planetary Exospheres. White paper submitted to Heliophysics Decadal Survey, [link](#).
- F. Hearty, S. Beland, J. Green, N. Cunningham, J. Barentine, M. Drosback, R. Valentine, A. Bondarenko, C. Schmidt, J. Walawender, C. Froning, J. Morse and P. Hartigan (2005) Colorado’s Near-Infrared Camera (AKA NIC-FPS) Commissioning on the ARC 3.5M Telescope, Proc. SPIE, Vol 5904, p. 199-211, [DOI](#).

Invited Colloquia

The Io Plasma Torus, Union College, Schenectady, NY, USA	2024
Io’s Atmosphere, Ionosphere, and Plasma Torus, LATMOS, FR	2023
Solar System Context in the Hunt for Exomoons, NASA JPL, Pasadena, CA, USA	2023

Alkali Emissions in the Lunar Atmosphere, UMASS, Lowell, MA, USA	2023
The Lunar Sodium Exosphere, Taiwan Space Union, Taiwan	2023
The Io-Jupiter Interaction, UMD, College Park, MD, USA	2022
Io's Atmosphere and Plasma Torus, Boise State University, Boise, ID, USA	2022
Optical Spectroscopy of Jupiter's Moons, AAVSO, Cambridge, MA, USA	2021
Observing the Exospheres of Mercury & the Moon, UMASS, Lowell, MA, USA	2020
Io's Escaping Atmosphere & Plasma Torus, Boston College, Boston, MA, USA	2018
Solar Transit Spectroscopy of Mercury's Exosphere, Universiteit van Amsterdam, NL	2018
Io's Escaping Atmosphere & Plasma Torus, Universität zu Köln, DE	2018
Io's Volcanic Atmosphere and Plasma Torus, Boston University, Boston, MA, USA	2018
Io's Plasma Torus Density & the S ⁺ Ribbon, Royal Institute of Technology, SE	2017
Small Telescope Applications: Mercury, Io & Comets, Université de Liège, BE	2017
Planetary Applications for Small Telescopes, Institute of Astronomy, Sofia, BG	2017
Visible Spectroscopy of the Io Plasma Torus, LESIA, l'Observatoire de Paris, FR	2016
Observations of Io, its Plasma Torus and Neutral Clouds, Lancaster Univ, UK	2016
Modern Planetary Applications for Small Telescopes, UMD, College Park, MD, USA	2015
Characteristics of Sodium Escape at Mercury, SERENA-HEWG, Killarney, IRL	2014
Atmospheric Escape in Our Solar System, Space Challenges, Sofia, BG	2013
Mercury's Sodium Atmosphere, AOSS, Univ. of Michigan, Ann Arbor, MI, USA	2012
Mercury's Tenuous Atmosphere, Heliophysics, NASA GSFC, Greenbelt, MD, USA	2012

Grants, Awards & Fellowships

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- James Webb Space Telescope Cycle 4 *Io's Auroral Emissions as a Tool to Investigate Atmosphere-Plasma Torus Interactions*, Co-I (PI Zac Milby, Caltech), TBD. JWST-GO-08857. Total budget: \$110,475. Funding to BU: TBD
 - Hubble Space Telescope Cycle 32 *A UV eclipse test for Callisto's atmosphere, plasma interaction and water ocean*, Co-I (PI Kurt Retherford, SwRI), 2025.10.01 to 2028.09.30. HST-GO-17739. Total budget: \$63,963. Funding to BU: \$5,000
 - NASA/NExSCI Keck Award *Joint Keck-Juno observations of Jupiter, its moons and its magnetosphere*, PI, 2024.08.01 to 2026.07.30. 80NSSC22K0954 (renewal). Total budget / funding to BU: \$120,000
 - Hubble Space Telescope Cycle 31 *HST-Juno Io Campaign: Connecting Volcanos to the Plasma Environment*, Co-I (PI Kurt Retherford, SwRI), 2023.12.01 to 2026.11.30. HST-GO-17470. Total budget: \$268,968. Funding to BU: \$13,160
 - NSF AST PAARE *Accessing the dark Arizona skies for research and education, a NCAT-BU partnership*, Co-PI (PI Dan Clemens, BU), 2023.08.24-2026.08.23. Total budget / funding to BU: \$256,696
 - NASA/NExSCI Keck Award *Recoil Heating in Mercury's Alkali Exotail*, PI (Science PI Patrick Lierle, BU), 2024.02.01 to 2025.01.31. Total budget / funding to BU: \$13,375.
 - NASA FINESST PI for Emma Lovett's Graduate Fellowship *Characterizing and Simulating Alkalis in Europa's Exosphere*, PI, 2023.09.01 to 2026.08.31. 22-PLANET22-0198. Total budget / funding to BU: \$150,000

- Hubble Space Telescope Cycle 30 *Ganymede's water atmosphere in eclipse*, PI, 2023.06.01 to 2026.05.31. HST-GO-17099. Total budget: \$75,414. Funding to BU: \$55,643
- NASA Discovery Data Analysis Program *Mercury's escaping sodium tail*, PI, 2022.09.15 to 2025.09.14. 80NSSC22K1303. Total budget: \$525,000. Funding to BU: \$393,997
- NASA/NEExSCI Keck Award *Joint Keck-Juno observations of Jupiter, its moons and its magnetosphere*, PI, 2022.08.01 to 2024.07.30. 80NSSC22K0954. Total budget / funding to BU: \$150,000
- NSF Astronomy and Astrophysics Research Grant. *Mass transport in Jupiter's magnetosphere: driven by internal or external processes?* Co-I/Institutional PI (PI Jeff Morgenthaler, Planetary Science Institute), 2021.09.01 to 2024.08.30. AST-2108416. Funding to BU: \$94,720
- NASA Solar System Observations *Dynamic Processes on the Galilean Satellites*, Co-I/Institutional PI (PI John Spencer, Southwest Research Institute), 2021.08.01 to 2024.07.31. 80NSSC21K1138. Funding to BU: \$59,981
- NASA Discovery Data Analysis Program *Investigating the Impactor Contribution to Mercury's Exosphere*, Co-I/Institutional PI (PI Aimee Merkel, Univ. Colorado), 2021.05.21 to 2024.04.30. 80NSSC21K1019. Total budget: \$551,469. Funding to BU: \$131,866
- NASA Science Mission Directorate *Characterizing Mercury's Exosphere with BepiColombo-PHEBUS: US-based Co-Investigators*, PI, 2020.10.13 to 2025.10.12, 80NSSC21K0051. Total budget / funding to BU: \$226,061
- NASA/NEExSCI Keck Award *Response of Io's atomic atmosphere and ionosphere to Jovian eclipse: joint observations with HIRES and HST*, PI, 2020.02.01 to 2020.09.30. 87/2020A-N079. Total budget / funding to BU: \$11,775
- NASA New Frontiers Data Analysis Program *The plasma distribution in the Io torus during the Juno epoch*, Co-I (PI Paul Withers, Boston Univ.), 2019.03.21 to 2022.02.28. 80NSSC19K0818. Total budget / funding to BU: \$289,272
- SOFIA Guest Observer Cycle 7 *Io's Atomic Sulfur Atmosphere in the Mid-IR*, PI, 2019.04.01 to 2020.03.31. 07-0221. Total budget / funding to BU: \$16,700
- NASA/NEExSCI NN-EXPLORE WIYN PI Data Award *Confirming a High Velocity Exo-Exosphere at HD 80606b*, PI, 2019.02.01 to 2021.01.31. N0223. Total budget / funding to BU: \$10,100
- NASA/NEExSCI Keck Award *Juno Support: Io's Auroral Emissions in Jovian Eclipse*, PI, 2019.02.01 to 2020.01.31. 84-208B-N110. Total budget / funding to BU: \$10,062
- Hubble Space Telescope Cycle 26 *Auroral and magnetospheric context for Juno in situ instruments during Cycle 26*, Co-I (PI Denis Grodent, Univ. Liege), 2019.03.01 to 2020.02.28. HST-GO-15638. Total budget / funding to BU: \$134,087
- NASA Solar System Workings *Physical Processes Governing Mercury's Alkali Exosphere*, PI, 2018.11.01 to 2021.03.31. 80NSSC19K0790. Total budget: \$352,275. Funding to BU: \$203,872

- NASA Solar System Observations *Ground-based observations of Mercury's exosphere in the post-MESSENGER era*, PI, 2018.03.01 to 2021.02.28. 80NSSC18K0857. Total budget: \$507,403. Funding to BU: \$165,281
- NASA Solar System Workings *The Ins and Outs of the Io Plasma Torus: understanding mass and energy transport using two decades of optical and radio observations*, Co-I (PI Jeff Morgenthaler, Planetary Science Institute), 2017.08.23 to 2020.08.22. 80NSSC17K0733. Total budget: \$526,604. Funding to BU: \$115,358
- Hubble Space Telescope Cycle 25 *Extreme Doppler Shifting of Io's Neutral Jets*, PI, 2018.03.01 to 2019.02.28. HST-GO-15147. Total budget: \$39,999. Funding to BU: \$28,006
- NSF Astronomy and Astrophysics Research Grant *The Influence of Mercury's Magnetosphere on Its Outermost Atmosphere*, Science PI (PI Luke Moore, Boston Univ.), 2016.07.15 to 2019.06.30. AST-1614903. Total budget / funding to BU: \$374,407
- NASA Earth and Space Sciences Fellowship *Mercury's Escaping Atmosphere*, Science PI (PI Michael Mendillo, Boston Univ.), 2010.03.15 to 2013.03.15. 10-Planet10F-0041. Total budget / funding to BU: \$90,000

Telescope Time Awarded

JWST, STScI (as Co-I, PI Z. Milby)	2025
IRTF, NASA (as Co-I, PI R. Cartwright)	2024
Hubble Space Telescope, STScI (as Co-I, PI L. Roth)	2024
Keck I & II (as Co-I, PIs P. Lierle and L. Moore)	2024, 2025
HST & JWST joint program, STScI (as Co-I, PI K. Retherford)	2023
WIYN, NASA NExScI	2022
Hubble Space Telescope, STScI (as US PI, PI L. Roth)	2022
Keck I & II, NASA NExScI (as PI & Co-I, PI L. Moore)	2022, 2023
Keck I & II, NASA NExScI (as Co-I, PIs L. Moore & K. de Kleer)	2021
Very Large Telescope, ESO (as Co-I, PI A. Oza)	2020
Keck I & II, NASA NExScI (as PI & Co-I, PI M. Vogt)	2020
THEMIS Solar Telescope, SOLARNET (as Co-I, PI V. Mangano)	2019, 2020
Big Bear Solar Observatory	2019
GREGOR Solar Telescope	2019
SOFIA, USRA	2019
IRTF, NASA (as Co-I, PI L. Moore)	2019
WIYN, NASA NExScI	2019
Keck I, NASA NExScI (as PI & Co-I, PI K. de Kleer)	2019
Hubble Space Telescope, STScI (as Co-I, PI D. Grodent)	2019
Hubble Space Telescope, STScI	2018
Dunn Solar Telescope, National Solar Observatory	2016
Vacuum Tube Telescope, SOLARNET	2016
GREGOR Solar Telescope, SOLARNET (as Co-I, PI V. Mangano)	2016
Very Large Telescope, ESO (as Co-I, PI B. Bonfond)	2015

Via Institutions Partnerships: Large Binocular Telescope, IRTF, Apache Point 3.5m, Lowell Discovery Telescope

Service (non-academic)

- At-large member of the IAU G5 Commission on Stellar & Planetary Atmospheres (2024 - 2027)
- Private sector advisor to the Group of Friends on Dark and Quiet Skies for the UN Committee on the Peaceful Uses of Outer Space (2024 -)
- Institutional Representative: Massachusetts Space Grant Consortium (2020 -)
- Journal Reviews: Planetary Science Journal, Icarus (outstanding reviewer award 2017), Journal of Geophysical Research, Geophysical Research Letters, Nature, Astronomy & Astrophysics
- Panelist for federal programs: 12x Research Opportunities in Space and Earth Science (ROSES) programs, Keck Time Allocation Committee, PDS Derived Data Review, Discovery Mission Extension Review, CNRS DIM-ORIGINES (France).
- Local Organizing Committees: Cool Stars 20 Conference, Boston University (2018), DPS Conference, Providence RI (2021)
- Scientific Organizing Committees: Jupiter Day, Boston University (2018)
- Session Chair: AGU: Dynamics of the Io-Jupiter System (2014) and Dynamic Exospheres of Terrestrial bodies through the Solar System (2023 & 2024), DPS/EPSC: Fire and Ice: Io and Beyond (2023), Io plasma torus splinter meetings at MOP (2017 & 2018), Exosphere/Magnetosphere, Mercury: Current and Future Science of the Innermost Planet, USRA (2018)

Team Activities & Memberships

- Instrument Science PI: Rapid Imaging Planetary Spectrograph: <http://carlschmidt.science/RIPS.html>
- Co-Investigator: ESA/JAXA BepiColombo mission PHEBUS instrument
- Private consultant: NASA SBIR Phase I to LambdaMetrics for development of the Serpentine Integrated Grating Spectrometer for Extreme Precision Radial Velocimetry
- International Space Science Institute Teams: The influence of Io on Jupiter's Magnetosphere (2016 - 2017), Surface Bounded Exospheres and Interactions in the Solar System (2020), Mass loss from Io's unique atmosphere: Do volcanoes really control Jupiter's magnetosphere? (2021 - 2022)
- Keck SCALES solar system science definition team
- Memberships: American Astronomical Society, International Astronomical Union, American Geophysical Union

Public Outreach

Subject Matter Expert, NASA Community College Network	2023 –
Volunteer Astronomer, High Rock Tower & Observatory, City of Lynn, MA	2022 –
Spectroscopy Lab Instructor, ISS Downlink Day, Boston University	2022
Lab Instructor, BU Academy Dept Visit, Boston Univ.	2022
Host, Navajo-Hopi Astronomy Outreach Program, Lowell Observatory	2018
Lecturer / Volunteer Astronomer, J.B. Coit Observatory, Boston Univ.	2017 –
Lecturer, Fan Mountain Observatory Public Nights, Univ. Virginia	2014 – 2015
Science Fair Judge, Virginia Piedmont Regionals, Charlottesville, VA	2014
Lecturer, McCormick Observatory Public Nights, Univ. Virginia	2013 – 2015
Lab Instructor, Upward Bound program, Boston Univ.	2010
Science Fair Judge, O'Bryant School for Math and Science, Roxbury, MA	2009
Workshop Coordinator, Sprout, www.thesprouts.org, Somerville, MA	2009 – 2013

Press & Media

NY Times: <i>Auroras Spotted on Neptune for the First Time, Lead to a New Mystery</i>	2025
Astronomy Magazine: <i>Scientists view Ganymede's aurorae dancing during eclipse</i>	2024
NY Times: <i>Europa, Thought to Be Habitable, May Be Oxygen-Starved</i>	2024
Astronomy Magazine: <i>Aurorae throughout our solar system and beyond</i>	2023
Live TV Interview, Hopewell Meteorite, Russian Television International (RTVi)	2023
AAS NOVA <i>Pizza Aurorae: Northern (and/or Southern) Lights on Io</i>	2023
NY Times: <i>New Auroras Found Glowing in the Skies of Jupiter's Moons</i>	2023
NY Times: <i>Telescopes Team Up to Forecast an Alien Storm on Titan</i>	2022
Boston Globe: <i>Bad weather may hurt viewing of rare lunar eclipse Friday in Mass.</i>	2021
Swedish National Public Television: <i>Today a storm from the moon pulls past the earth</i>	2021
NY Times: <i>The Moon Has a Comet-Like Tail</i>	2021
Wall Street Journal: <i>Comet Neowise as Seen Around the World</i>	2020
Sky & Telescope: <i>Comet NEOWISE Dazzles at Dusk</i>	2020
Fox News: <i>Comet NEOWISE may have sodium tail, new images suggest</i>	2020
TV Interview, <i>Space Challenges Documentary</i> , Bulgarian National Public Television	2017
TV Interview, NASA ScienceCast: <i>The 2016 Transit of Mercury</i>	2016
Content Advisor, Science in the News, Harvard University GSAS	2013 - 2016
Radio Interview, Science Straight Up, WTJU FM	2014
Phys.org: <i>Mercury's comet-like appearance spotted by satellites looking at the Sun</i>	2010
Universe Today: <i>STEREO Catches Mercury Acting Like a Comet</i>	2010